

CLINICAL PRACTICE UPDATES

AGA Clinical Practice Update on Diagnosis and Treatment of Hemorrhoids: Expert Review



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DESCRIPTION: Hemorrhoids are a common problem. The diagnosis and treatment can be challenging. Gastroenterologists have much to offer these patients. The purpose of this American Gastroenterological Association (AGA) Clinical Practice Update Expert Review is to provide best practice advice (BPA) covering the diagnosis and treatment of hemorrhoid disease.

METHODS: This expert review was commissioned and approved by the AGA Institute Clinical Practice Updates Committee (CPUC) and the AGA Governing Board to provide timely guidance on a topic of high clinical importance to the AGA membership and underwent internal peer review by the CPUC and external peer review through standard procedures of Clinical Gastroenterology and Hepatology. These BPA statements were drawn from a review of the published literature and from expert opinion. Because systematic reviews were not performed, these BPA statements do not carry formal ratings regarding the quality of evidence or strength of the presented considerations.

BEST PRACTICE ADVICE STATEMENTS

- BPA 1:** The diagnosis and treatment of hemorrhoids is within the purview of the gastroenterologist. The diagnosis and grading of hemorrhoids is easily made by taking a history from the patient and examining the patient. Symptoms caused by hemorrhoids include bleeding, itching, discomfort, and/or prolapse. Hemorrhoids only cause significant pain when acutely thrombosed. Sharp pain on defecation is most likely anal fissure.
- BPA 2:** Dietary and lifestyle modifications, including increasing fiber intake and avoiding straining or prolonged time on the toilet, are reasonable first-line therapies for symptomatic hemorrhoids. The use of sitz baths for symptom improvement in symptomatic hemorrhoids is often advised, but scientific data is limited.
- BPA 3:** Topical treatments, including anesthetics, astringents (witch hazel), corticosteroids, and vasoactive agents, can be considered for treatment of symptomatic hemorrhoids, but there is little data to support efficacy. Topical steroids should not be used for more than 2 weeks at a time.
- BPA 4:** Anoscopy should be performed, whenever possible, on every new patient with suspected hemorrhoids, prior to treatment, to ensure accurate diagnosis.
- BPA 5:** Both hemorrhoid banding and infrared coagulation are safe, effective, and easy to perform in the office setting. Infrared coagulation and rubber band ligation have similar benefits in the short term. Rubber band ligation has longer-term benefits for treatment of prolapsing hemorrhoids and recurrent bleeding. Hemorrhoid banding or infrared coagulation should be employed prior to surgical hemorrhoidectomy for grades 1 to 3 hemorrhoids.

- BPA 6:** As part of informed consent for hemorrhoid therapies, the patient must be made aware of the small possibility of pelvic sepsis as a complication. Patients should be counseled about the risk and instructed to present to the emergency department immediately for evaluation, if indicated.
- BPA 7:** In patients with active Crohn's disease or ulcerative colitis, hemorrhoid disease management should be delayed until complete remission is achieved.
- BPA 8:** Hemorrhoids occur in up to two-thirds of women during pregnancy. Treatment should generally involve conservative management, including fiber, treatment of constipation, and topical ointments. If symptoms persist postpartum, or if a woman is planning further pregnancies, standard treatment such as banding or infrared coagulation can be considered.
- BPA 9:** Acute thrombosed hemorrhoids are often extremely painful. They are best treated surgically with incision and drainage.
- BPA 10:** Consultation with a surgeon should be offered to patients with grade 3 internal hemorrhoids who fail banding procedures or have associated external hemorrhoids. Large skin tags can be removed without a hemorrhoidectomy if they are not associated with significant hemorrhoids. Grade 4 internal hemorrhoids require surgical hemorrhoidectomy.
- BPA 11:** Patients with cirrhosis and hemorrhoids should be carefully examined so as not to confuse hemorrhoids with rectal varices. Hemorrhoids in patients with cirrhosis can be treated with banding or infrared coagulation. In patients with significant coagulopathy, infrared coagulation is preferred to banding. Concomitant portal hypertension should not alter this approach. For most clinicians, significant coagulopathy means a platelet count of less than 50,000 per microliter or international normalized ratio greater than 2.0. The presence of concomitant portal hypertension should not alter this approach.

Hemorrhoids are a common concern for patients, and gastroenterologists frequently care for patients with hemorrhoidal complaints. There are a wide range of treatment options ranging from behavioral and dietary modifications to office-based procedures or surgical therapy. Selection of an appropriate treatment path may be complex and is guided by patient symptom(s) and severity of disease. Navigating these treatment modalities, in an evidence-based approach, is imperative to appropriately treat and manage symptomatic hemorrhoids.

BPA 1: The diagnosis and treatment of hemorrhoids is within the purview of the gastroenterologist. The diagnosis and grading of hemorrhoids is easily made by taking a history from the patient and examining the patient.

Symptomatic hemorrhoids are common, with almost one-half the population of 50-year-olds having symptoms of active hemorrhoid disease. Over 23 million Americans suffer from hemorrhoid symptoms, resulting in about 3.5 million visits to the doctor and an expenditure of over \$500 million in health care costs.^{1,2} Most hemorrhoid disease, just like most anorectal disease, can be diagnosed and managed in the office and is increasingly managed by the gastroenterologist. With age, enlargement and displacement of the normal vascular cushions in the anal canal can lead to symptoms. Hemorrhoid symptoms include bleeding, itching, discomfort, prolapse, or a combination of these.

Hemorrhoids only cause significant pain when acutely thrombosed.

Internal hemorrhoids are graded according to the degree of prolapse (Table 1). This grading system, also known as Goligher's classification, is based mostly on the patient's history. Although the findings can be confirmed on examination, a good history will classify the severity of the internal hemorrhoids under this classification.³ There is no standard classification for external hemorrhoids, and the description is frequently subjective. Typically, rectal bleeding is more likely from internal hemorrhoids and itching more likely from external hemorrhoids. If the patient complains of sharp pain on defecation or has sharp pain on digital rectal exam, then the diagnosis is most likely anal fissure. However, anal fissure and hemorrhoid disease may coexist in up to 20% of cases. In this situation, the anal fissure should be treated first.⁴ During inspection, the perianal area may reveal redundant skin tags, some with a bulge. These are often vestiges of previous thrombosed hemorrhoids and are harmless and best left alone. Clinically, it is important to differentiate rectal prolapse from prolapsing hemorrhoids. Rectal prolapse is the intussusception of the rectal wall through the anal canal, presenting with circular folds of pink rectal mucosa, because of loss of normal rectal attachments. Prolapsed hemorrhoids should have radial folds and engorged blood vessels.

Table 1. Goligher's Classification of Hemorrhoid Disease

Grade	Description
Grade 1	Do not prolapse below the dentate line; bleeding only
Grade 2	Prolapse below the dentate line, during defecation but spontaneously reduce
Grade 3	Prolapse below the dentate line and require manual reduction
Grade 4	Prolapse and stay below the dentate line and are not reducible

Although the diagnosis is frequently made on the basis of history alone, a good anorectal examination in the left lateral decubitus position including a perianal examination, a careful digital rectal exam, and if available, anoscopy, aids diagnosis. Anoscopy allows a good view of the entire anal canal so that condylomas or early anal cancer is not missed. A full colonoscopy should be considered in patients with rectal bleeding to rule out concomitant or more serious disease.

BPA 2: Dietary and lifestyle modifications, including increasing fiber intake and avoiding straining or prolonged time on the toilet are reasonable first-line therapies for symptomatic hemorrhoids. The use of sitz baths for symptom improvement in symptomatic hemorrhoids is often advised, but scientific data is limited.

Constipation has long been associated with hemorrhoid development. A study using colonoscopy data from a large clinical trial found that constipation, including straining during bowel movements, was associated with increased risk of hemorrhoids.⁵ This study also found that a diet high in grain fiber was associated with a reduced risk for hemorrhoids.⁵ Measures to improve constipation, specifically increasing fiber intake, are recommended in many guidelines on symptomatic hemorrhoid management.^{6,7} Evidence for this primarily comes from 2 systematic reviews analyzing 7 studies comparing fiber with placebo, and found moderate evidence to support the use of fiber to improve overall symptoms and specifically bleeding in patients with symptomatic hemorrhoids.^{8,9}

A study of 102 patients with advanced, symptomatic hemorrhoids addressed altered defecation patterns including increased straining, prolonged defecation time, frequent bowel movements, and fiber intake. In this study, patients were advised to limit time on the toilet to 3 minutes, defecate once daily, avoid straining, and consume at least 20 to 30 grams of fiber. Results showed improvement in prolapse and bleeding.¹⁰

The use of sitz baths to treat symptomatic hemorrhoids is not addressed in several of the most recent guidelines from Gastrointestinal and Colorectal Surgery societies.^{6,7} Sitz baths are used to treat several anorectal disorders, but there is limited scientific evidence for symptomatic hemorrhoids.¹¹⁻¹³ Manometric studies have shown that patients with hemorrhoids have increased anal sphincter pressures.¹⁴ It is postulated

that warm sitz baths reduce anal sphincter pressure and may improve symptoms in hemorrhoid disease. A review of 36 articles on the use of sitz baths and anorectal disorders found little evidence to support improvement in symptoms and highlighted the need for more research in this area.¹⁵ Moreover, patients may feel that the use of bidets may limit puritis related to hemorrhoid disease by improving hygiene and avoiding repeated friction from toilet paper.

BPA 3: Topical treatments, including anesthetics, astringents (witch hazel), corticosteroids, and vasoactive agents can be considered for treatment of symptomatic hemorrhoids, but there is little data to support their efficacy. Topical steroids should not be used for more than 2 weeks at a time, because prolonged use can thin the skin and cause it to become more sensitive.

There are many over-the-counter options for the treatment of symptomatic hemorrhoids. These specifically are classified into anesthetics, astringents and protectants, corticosteroids, and vasoactive agents. Some provide temporary relief of symptoms, particularly during a flareup, for example, following a bout of constipation, although their efficacy has not been well-studied.^{6,7,11-13} For symptoms of burning and soreness, ointments that contain lidocaine pramoxine or benzocaine might be helpful, whereas inflammation and swelling with intense itching, may respond better to a brief course of hydrocortisone cream or suppository. For minor itching and irritation, witch hazel is helpful. Some Preparation H products have phenylephrine, a vasoconstrictor that will temporally shrink blood vessels and reduce swelling or bleeding. Compounds with zinc oxide, mineral oil, or petrolatum provide protective barriers that can help heal and soothe the irritated skin around the hemorrhoids.

BPA 4: Anoscopy should be performed, whenever possible, on every new patient with suspected hemorrhoids prior to treatment to ensure accurate diagnosis.

Anoscopy is superior to flexible endoscopy and is the preferred method to examine the anal canal (Figure 1).¹⁶ Anoscopy should not be confused with high-resolution anoscopy, which involves the use of a colposcope meant to detect intraepithelial neoplasia or early changes related to anal cancer. Anoscopy is done following the digital rectal exam with the patient in the

left lateral decubitus position. The procedure is easy to learn and perform.¹⁷ The best anoscopes to examine the anal canal and visualize hemorrhoids, are disposable slotted or beveled anoscopes with a built-in light. Beveled scopes can be rotated, without risk of trauma to the hemorrhoid column (Figure 2 and Supplementary Video 1). In tense or nervous patients, the anoscope may be easier to insert by asking the patient to bear down during the process, which relaxes the pelvic floor muscles. Although anoscopy is usually done in the office setting, it can certainly be performed following a colonoscopy for better assessment of the anal canal.

BPA 5: Both hemorrhoid banding and infrared coagulation are safe, effective, and easy to perform in the office setting. Infrared coagulation and rubber band ligation have similar benefits in the short term. Rubber band ligation has longer-term benefits for the treatment of prolapse and recurrent bleeding. Hemorrhoid banding or infrared coagulation should be employed prior to surgical hemorrhoidectomy for grades 1 to 3 hemorrhoids.

Rubber band ligation of internal hemorrhoids is offered as an office-based procedure and can be effective treatment of grades 1 to 3 hemorrhoids (Supplementary Video 2). The procedure is performed

in the office without sedation. The lubricated anoscope is inserted into the anus, and the internal hemorrhoid tissue is identified. The rubber band is placed above or proximal to the dentate line and is therefore well-tolerated. The ligated hemorrhoidal tissue will undergo ischemia and necrosis, leaving behind an ulcer that eventually will scar and tack the mucosa to the rectal wall. This achieves 2 purposes: (1) to prevent mucosal prolapse; and (2) to reduce blood flow into the hemorrhoid and cause it to shrink. Rubber band ligation has been found to be the most effective office-based procedure, with success rates ranging between 66% and 94%.¹⁸ Success is generally defined as resolution of symptoms and successful treatment of prolapse. There are several hemorrhoid banding devices on the market (Figure 3). Although banding is effective for grades 1 to 3 hemorrhoids, efficacy of treatment varies by grade. A Cochrane review identified that rubber band ligation was found to be equally effective as surgical excision for the treatment of grade 2 hemorrhoids, but not as effective as surgical excision for grade 3 hemorrhoids.¹⁹ Complications occur in 2% of cases and range from urinary retention and bleeding to secondary fissure formation. If a band is deployed too close to the dentate line, it will cause pain and should be removed immediately, easily accomplished by the clinician using the finger to roll the band off of the hemorrhoid or directly massaging the hemorrhoid in a circular fashion. Although extremely rare, pelvic sepsis has been reported as a possible complication of rubber band ligation.²⁰

Infrared coagulation (IRC) uses heat energy to induce coagulation and fibrosis in the submucosal layer around the origin or pedicle of the internal hemorrhoid to reduce blood supply and cause the hemorrhoid to shrink (Supplementary Video 3). The procedure is commonly performed under direct vision through an anoscope. It works particularly well for grades 1 and 2 internal hemorrhoids and also for many grade 3 hemorrhoids.

The IRC 2100 (Redfield Corp) is a nonendoscopic system consisting of a compact power unit with a tungsten-halogen lamp (Figure 4). A pistol-shaped hand-held applicator connects to the power unit and delivers light energy to the tip of the shaft. The energy is applied in 1- to 1.5-second bursts just proximal to the visible internal hemorrhoid column in 5 or 6 spots. If the patient feels pain or burning during the procedure, a transmural burn might be occurring, and the energy applied needs to be reduced. This is repeated for each column, resulting in coagulation and necrosis, visible as a white spot. The treatment may need to be repeated in a few weeks to completely control the symptoms. Occasional post-procedure discomfort can be treated with over-the-counter analgesia. Although IRC is a safe and rapid procedure, the risks include pain and burning if too much energy is applied, causing a transmural burn or ulceration. IRC is not suitable for large, prolapsing hemorrhoids.

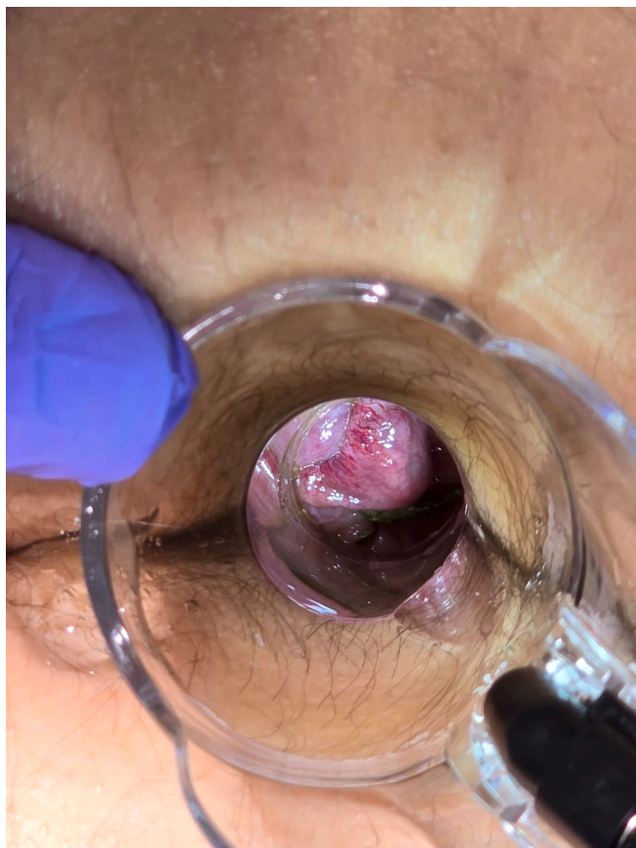


Figure 1. This is a picture of the right posterior hemorrhoid column as seen through an anoscope. The changing color is the junction between the internal and external hemorrhoid, the so-called dentate line.



Figure 2. This photograph shows a beveled, disposable plastic anoscope with a built-in light. The built-in light makes it very convenient and easy to use. The second anoscope in the white background is a slotted anoscope, and therefore cannot be rotated while inside the anal canal so as not to cause damage to the internal hemorrhoid column.

Rubber band ligation and IRC seem to have similar outcomes in the short term. However, studies suggest that, although rubber band ligation may be associated with more post-procedure discomfort, its efficacy is longer lasting than IRC.^{20–23} Sclerotherapy for the treatment of hemorrhoids has been largely abandoned because of the potential for complications, such as a misplaced sclerosing injection and reduced efficacy compared with safer modalities.²⁴ IRC is favored in certain circumstances, such as in patients who are on anticoagulants or antiplatelet agents, where there is an increased risk of bleeding when the band falls off. IRC may also be preferred in pregnancy and in young people with significant irritable bowel syndrome symptoms. Lower abdominal cramping that could follow hemorrhoid banding can cause undue anxiety in the pregnant patient. Patients with IBS tend to be more sensitive to banding, and therefore IRC may be a better option, but there is very little evidence in the literature to support this, so further studies are needed. Any post-procedural discomfort can be treated with over-the-counter analgesia, such as acetaminophen or nonsteroidal anti-inflammatory drugs.



Figure 3. There are several different types of disposable hemorrhoid banding devices on the market. Here is a figure showing 3 such devices. Although they look different, they all have the same mechanism of action in that a syringe-like suction device pulls the hemorrhoid mucosa into a cylinder and releases a band over the mucosa. They are not difficult to learn to use but do need some practice.

BPA 6: As part of informed consent for hemorrhoid therapies, the patient must be made aware of the small possibility of pelvic sepsis as a complication. Patients should be counseled about the risk and instructed to present to the emergency department immediately for evaluation, if indicated.



Figure 4. This is a photograph of the infrared coagulation device, where the top black handle has a probe with a tip at the end that becomes extremely hot to provide diathermy to the proximal end of the hemorrhoid column. This procedure is performed under direct vision through an anoscope and is easy to learn.

Possible complications following hemorrhoid banding include bleeding, pain, vasovagal symptoms, slippage of bands, priapism, and thrombosed hemorrhoids. Major complications include pelvic sepsis and death, which are fortunately very rare. In one prospective study of 512 patients who underwent rubber band ligation, 37 patients or 7.2%, had complications, 4.7% of these were minor.¹⁸ Significant bleeding, severe pain, or perianal fistula developed in 2.5% of the patients.¹⁸ Infectious complications following rubber band ligation include pelvic sepsis, which is the most serious and can be lethal, liver abscesses, Fournier's gangrene, tetanus, and bacterial endocarditis. Pelvic sepsis can develop any time between 3 and 10 days following banding. Although rare, pelvic sepsis is a potentially life-threatening complication that can develop after hemorrhoid surgery and office-based procedures such as rubber band ligation. Patients may present with severe post-procedural pelvic pain, urinary dysfunction, and worsening swelling and induration of the perineum. Patients may also present with systemic symptoms such as fevers, muscle aches, and changes in mental status. Presentation of any concerning symptoms should prompt an immediate evaluation. The most frequent symptoms from pelvic sepsis are severe pain and urinary retention. We provide written instructions to all patients following hemorrhoid banding to return to the emergency room immediately should they develop any of the following:

severe pain, urinary retention, fever, chills, or significant bleeding. A diagnosis of pelvic sepsis requires immediate and aggressive intervention, which may or may not require surgery. Although complications with IRC are less common, aggressive application of thermal energy could cause a transmural burn that could result in an abscess.

BPA 7: In patients with active Crohn's disease or ulcerative colitis, hemorrhoid disease management should be delayed until complete remission is achieved.

There is very little in the scientific literature about the safety and efficacy of hemorrhoid treatment in the presence of inflammatory bowel disease (IBD). Consultation with an IBD expert, and a surgeon are often required prior to hemorrhoid treatment. In patients who go on to surgical hemorrhoidectomy, there is a higher risk of complications in patients with IBD, although this risk seems to be lower in more recent studies, possibly due to the introduction of biologics.²⁵ The literature is scant and based on small series, regarding the outcome of office management of hemorrhoid disease in IBD. Expert opinion would dictate that IBD is treated aggressively, and the patient should be in remission prior to hemorrhoid banding or IRC. Should conservative management be unsuccessful, then surgical options should be explored in highly selective cases.²⁶ In one retrospective study of excisional hemorrhoidectomy in 36 patients with Crohn's disease in remission, 4 patients developed complications over a mean follow-up of 31 months that included 1 anal stricture, 1 perianal abscess, 1 nonhealing wound, and 1 hemorrhoid recurrence.²⁷ Anal skin tags are frequently seen as a result of long-standing external hemorrhoid disease but also occur frequently in IBD, particularly in Crohn's disease. In either situation, surgery is not advised, although young patients presenting with abdominal symptoms of pain and or diarrhea need to be worked up to exclude Crohn's disease.²⁸ A thorough anoscopic exam and sometimes examination under anesthesia may be necessary to exclude active disease and provide an accurate diagnosis.

BPA 8: Hemorrhoids occur in up to two-thirds of women during pregnancy. Treatment should generally involve conservative management, including fiber, treatment of constipation, and topical ointments. If symptoms persist postpartum, or if a woman is planning further pregnancies, standard treatments such as banding or infrared coagulation can be considered.

Hemorrhoids usually develop during the third trimester of pregnancy, largely from the enlarging uterus increasing the intra-abdominal pressure and causing venous stasis and vascular engorgement of the inferior mesenteric veins.²⁹ Also, extrinsic compression of the rectum may cause excessive straining and prolonged defecation time leading to constipation. Additionally, prolonged pushing especially during the second

stage of labor may induce or aggravate hemorrhoids and cause prolapse.²⁹ Hemorrhoids are very common during pregnancy, with one study of 835 pregnant women reporting a prevalence of 86%.^{29,30} Another study found that 33% had thrombosed external hemorrhoids or anal fissures during pregnancy and/or postpartum.³¹ A prospective cohort study of 94 patients that followed women from the second trimester up to 3 months postpartum found that anal symptoms were reported by 50% during pregnancy, 14% developed hemorrhoidal prolapse during the third trimester, 14.6% had hemorrhoid thrombosis, and 13.5% developed prolapsed hemorrhoids in the immediate postpartum at 3 months.³² Another prospective cohort study that evaluated 290 women at 4 time points found that 40.7% developed hemorrhoids in the first trimester, and 61% in the third trimester, and 37.4% during or after delivery.³¹ Multivariate analysis identified specific risk factors that included excessive straining during delivery for more than 20 minutes (odds ratio [OR], 29.75%; 95% confidence interval [CI], 4.00–221.23), birthweight of newborn >3800 g (OR, 17.99; 95% CI, 3.29–98.49), constipation (OR, 18.98; 95% CI, 7.13–50.54), and personal history of perianal diseases (OR, 11.93; 95% CI, 2.18–65.30).

Treatment of hemorrhoids in pregnancy primarily involves the use of topical medications including hydrocortisone suppositories to reduce inflammation, bleeding, and/or pain with banding or IRC delayed until postpartum. Systemic absorption from these ointments or creams (see BPA 3) is minimal, and they are therefore considered safe in pregnancy. When necessary, symptomatic hemorrhoids can be managed mostly in an office setting in a similar fashion as nonpregnant patients are managed if one cannot wait until after delivery.³³ During pregnancy, IRC may be preferred to hemorrhoid banding because there is less post-procedural discomfort. It is also advisable to involve the obstetrician regarding any treatment decisions. There are few randomized studies of hemorrhoids in pregnancy, and hence, most treatment recommendations are based on expert opinion.²⁹

BPA 9: Acute thrombosed hemorrhoids are often extremely painful. They are best treated surgically with incision and drainage.

Thrombus formation within the external hemorrhoid can cause significant swelling and pain. Patients will complain of feeling a mass or a nodule, and this can be associated with bleeding. A thorough evaluation is important to diagnose a thrombosed external hemorrhoid. Historically, when a patient presents early in the course of disease (within 72 hours), excision in the office or operating room is offered. Early incision and drainage (I&D) provides the most effective and rapid relief of symptoms compared with conservative measures. When I&D is performed, it is important to “de-roof” the cavity so that it does not reform with the incision closing over. It is important to distinguish between I&D, which can be performed by a gastroenterologist in the office or done in the emergency room,

and thrombectomy, which is slightly more involved, involving the removal of the entire thrombosed part of the hemorrhoid and using sutures to close, which is usually performed by surgeons. When it comes to immediate relief of pain, the outcomes are very similar. In office, I&D of a thrombosed hemorrhoid is very easy to perform and is shown in the accompanying video ([Supplementary Video 4](#)). Antibiotics are not required following this procedure.

BPA 10: Consultation with a surgeon should be offered to patients with grade 3 internal hemorrhoids who fail banding procedures or have associated external hemorrhoids. Large skin tags can be removed without a hemorrhoidectomy if they are not associated with significant hemorrhoids. Grade 4 internal hemorrhoids require surgical hemorrhoidectomy.

Surgical opinion should be sought for patients who remain symptomatic despite medical therapy. Most symptomatic grade 1, grade 2, and many grade 3 internal hemorrhoids can be treated by office-based procedures. Patients with symptomatic grades 1 to 3 hemorrhoids who cannot tolerate or fail office-based procedures, or those with concomitant external hemorrhoids, should be referred to surgery. Also, patients with grade 3 hemorrhoids who do not respond well to banding and all patients with grade 4 hemorrhoids should be referred to surgery.

Other surgical options for internal hemorrhoids include hemorrhoidopexy and Doppler-guided hemorrhoidectomy. A circular stapling device is used in surgical hemorrhoidopexy to plicate the mucosa by creating a mucosal anastomosis. This technique was found to be well-tolerated, with patients experiencing less postoperative pain compared with excision hemorrhoidectomy.³⁴ However, on long-term follow-up, patients were found to have an increased rate of recurrence. Due to the increased risk for significant complications, including rectovaginal fistula, rectal perforation, bleeding and staple-line stricture, hemorrhoidopexy has fallen out of favor as a surgical option for the treatment of internal hemorrhoids.³⁵

Doppler-guided hemorrhoidectomy utilizes a Doppler probe to identify and ligate the hemorrhoid artery without an excision. In some instances, the surgeon may choose to perform a mucosal pexy for symptomatic, prolapsed hemorrhoids. As no excision is performed, patients report less postoperative pain and it is well-tolerated with minimal complications.³⁶

Surgical excision of hemorrhoids remains the definitive procedure for surgical management of hemorrhoids.³³ In the surgical suite, the entire hemorrhoid complex (internal and external) is dissected off the internal anal sphincter muscle and excised. The surgeon may choose to leave the incision site open or close it with suture; however, the closed approach is associated with a decreased risk for postoperative bleeding and faster wound healing. The overall complication rate after surgical excision is

low—between 1% and 2%. Complications range from urinary retention, bleeding, infection and fistula formation to the rare but morbid complication of pelvic sepsis.

Patients with external hemorrhoids can be offered surgical excision if they present soon after the onset of symptoms. Most thrombosed external hemorrhoids will resolve without any surgical treatment, but excision can provide early relief of symptoms. If surgery is offered, complete excision and evacuation of the clot should be performed.

BPA 11: Patients with cirrhosis and hemorrhoids should be carefully examined so as not to confuse hemorrhoids with rectal varices. Hemorrhoids in patients with cirrhosis can be treated with banding or infrared coagulation. In patients with significant coagulopathy, infrared coagulation is preferred to banding. Concomitant portal hypertension should not alter this approach. For most clinicians, significant coagulopathy means a platelet count of less than 50,000 per microliter or international normalized ratio greater than 2.0.

Hemorrhoid disease is commonly found in the cirrhotic patient. Rectal varices are portosystemic collaterals that develop as a complication of portal hypertension. Cirrhotic patients with portal hypertension can experience exacerbation of hemorrhoidal vascular congestion and have rectal varices in 30% of cases.³⁷ It is important to differentiate rectal varices from hemorrhoids. Rectal varices tend to be found in the mid and more proximal rectum, compared with hemorrhoids, which are usually distal and go across the dentate line. Although it is uncommon for rectal varices to bleed, the bleeding can be life-threatening. Although endoscopic treatment with cyanoacrylate injection or with coils frequently controls bleeding, the risk of embolism remains and rebleeding rates are high. Management of underlying portal hypertension is essential in patients with bleeding rectal varices, with procedures such as transhepatic portosystemic shunts, or as an alternative, balloon-occluded, retrograde transvenous obliteration by interventional radiology and are often effective. One study found this technique to be 100% successful, with a 72% clinical success rate at 1-month follow-up.³⁸

Conclusion

Hemorrhoid disease is a common cause of anorectal complaints, such as rectal bleeding, perianal pain, and discomfort. The diagnosis and initial management of hemorrhoids can be performed by the gastroenterologist. Anoscopy and digital rectal examination are essential parts of the anorectal examination and can be performed in the outpatient clinic. Hemorrhoid banding and IRC are safe and effective modalities for the management of grade 1 and grade 2 hemorrhoids, and the decision to refer to a surgeon should be individualized. Special consideration should be placed for patients who are pregnant and patients with cirrhosis. Surgical hemorrhoidectomy can be a low-risk procedure, but patients

should be counseled on the possibility of complications such as bleeding, recurrence, and pelvic sepsis.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Clinical Gastroenterology and Hepatology* at www.cghjournal.org, and at <https://doi.org/10.1016/j.cgh.2026.04.008>.

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All authors contributed in writing the updates and corrections.

Conflicts of interest

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